



Employee Attrition Analysis and Prediction

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WHY IS EMPLOYEE ATTRITION IMPORTANT?

High Costs: Replacing a worker costs significantly more than keeping one.

Lost Knowledge: Critical "know-how" and skills walk out the door.

Lower Productivity: Teams slow down while waiting for and training new hires.

Team Burnout: Remaining staff get overworked, leading to lower morale.

Damaged Reputation: High turnover makes it harder to hire great people.

Customer Impact: Inconsistent service leads to unhappy clients.



Project Overview: Addressing Employee Attrition

The Problem: High turnover increases costs and disrupts team productivity.

The Goal: Identify why employees leave and predict who might be at risk.

The Method: Analyze workforce data to find the "key drivers" of attrition.

The Outcome: Build predictive models to help management take action *before* people quit.

The Value: Shift from reactive hiring to proactive employee retention.

Pillar 1: Identify At-Risk Employees



Recognition & Engagement Gaps

Monitor engagement scores and recognition programs



Performance Trends

Track performance changes and career progression signals



Exit Interviews & Feedback

Implement targeted strategies and interventions

Pillar 2: Cost Optimization



Reduce Recruitment Costs

Improve time-to-hire through internal mobility and succession planning



Streamline Onboarding

Accelerate productivity with structured training programs



Minimize Attrition Impact

Retain top performers through career development and benefits

Pillar 3: Workforce Planning



Predictive Analytics

Use data-driven insights to forecast turnover and skill gaps



Strategic Alignment

Align retention strategies with business goals and growth plans



Employee Satisfaction

Improve engagement through career development and culture



ML Workflow: End-to-End Pipeline

Phase 1: Data Collection & Preprocessing



Data Sources

HR systems, payroll, performance records, employee demographics



Data Cleaning

Handle missing values, remove duplicates, fix outliers and inconsistencies



Data Preprocessing

Normalize numerical data, encode categorical variables, balance classes

Phase 2: Exploratory Data Analysis (EDA)



Pattern Discovery

Identify attrition trends, distribution analysis, statistical summaries



Correlation Analysis

Analyze relationships between features and attrition, identify key drivers



Visualization

Create heatmaps, histograms, scatter plots to understand data patterns

Phase 3: Feature Engineering



Feature Creation

Tenure categories, engagement scores, work-life balance metrics



Feature Transformation

Scaling, normalization, log transformations, polynomial features



Feature Selection

Select most relevant features using correlation and importance analysis

Phase 4: Model Building & Evaluation



Model Development

Logistic Regression, Decision Trees, Random Forests, Gradient Boosting



Model Evaluation

Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix



Hyperparameter Tuning

Optimize model performance using cross-validation and grid search

Phase 5: Actionable Insights & Visualization



Streamlit Dashboard

Interactive visualizations for attrition trends, at-risk employees, department analysis



Business Insights

Identify high-risk employee segments and actionable retention strategies



HR Recommendations

Targeted interventions, career development, compensation adjustments

Streamlit Dashboard Components

Attrition Rate by Department - Filter and compare attrition across teams

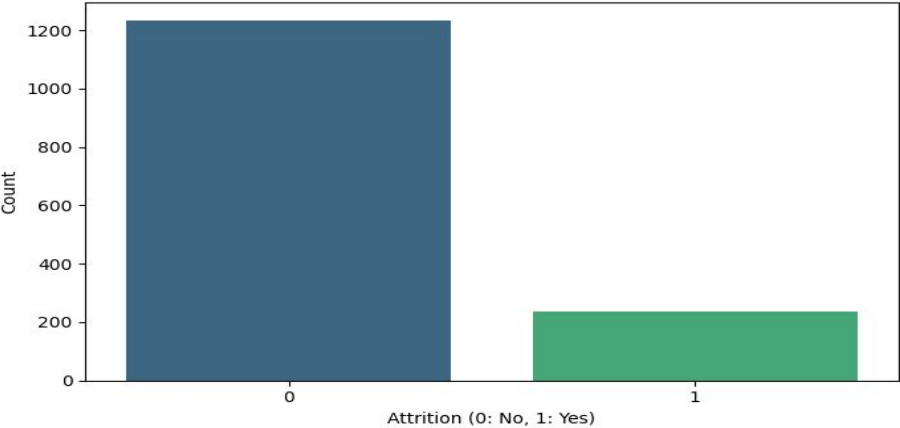
Risk Prediction Widget - Enter employee details to get attrition probability

Salary vs Satisfaction Analysis - Visualize compensation impact on retention

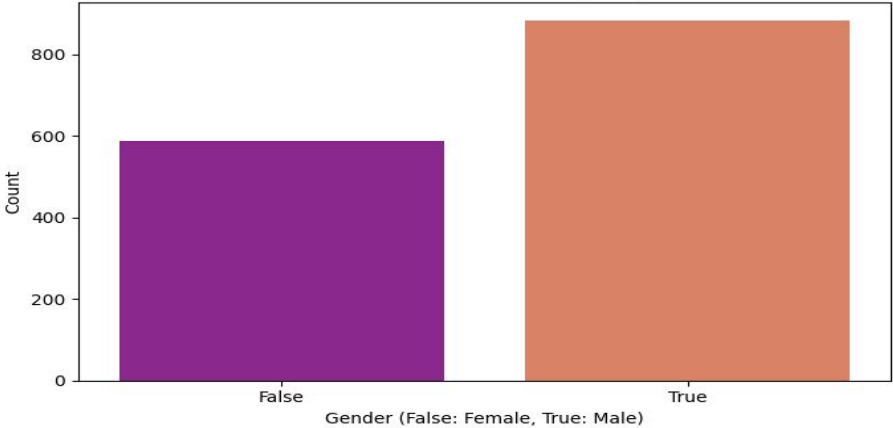
Tenure Trends - Analyze retention by experience level

Distribution of Key Categorical Features

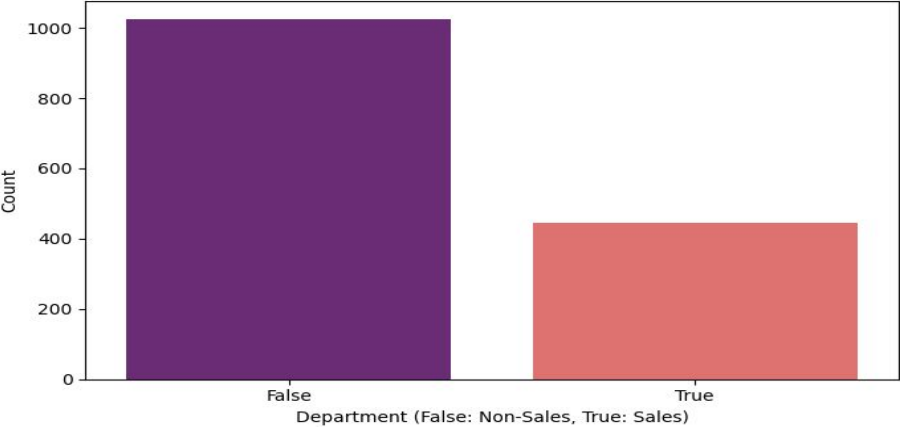
Distribution of Attrition



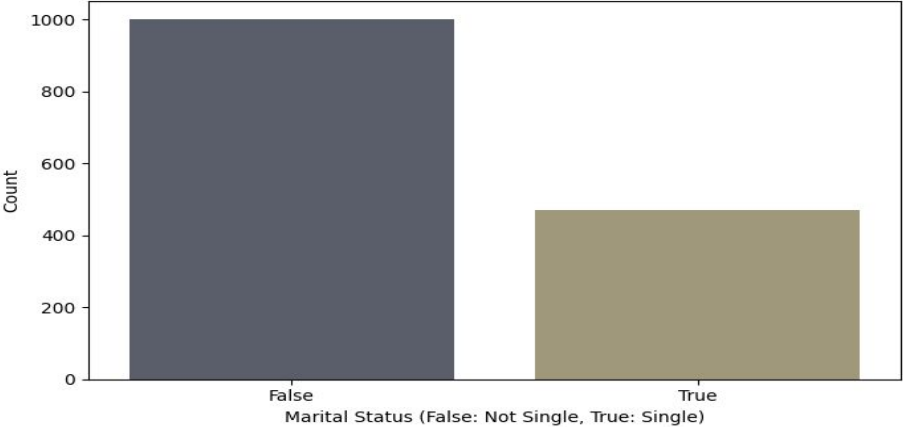
Distribution of Gender (Male)



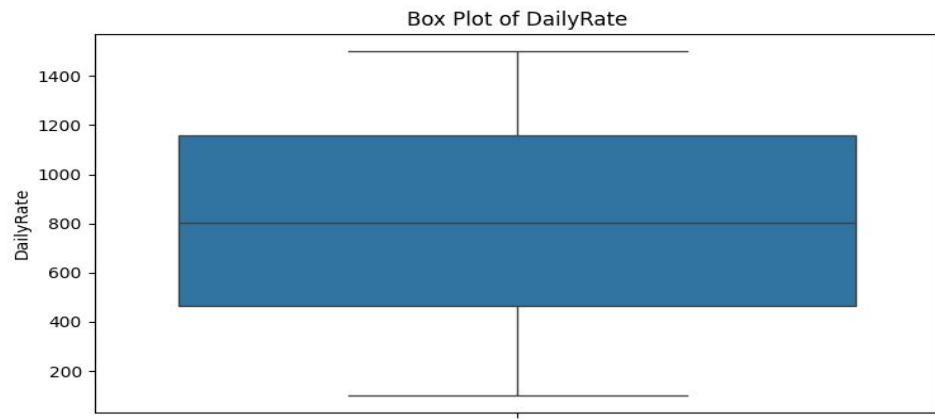
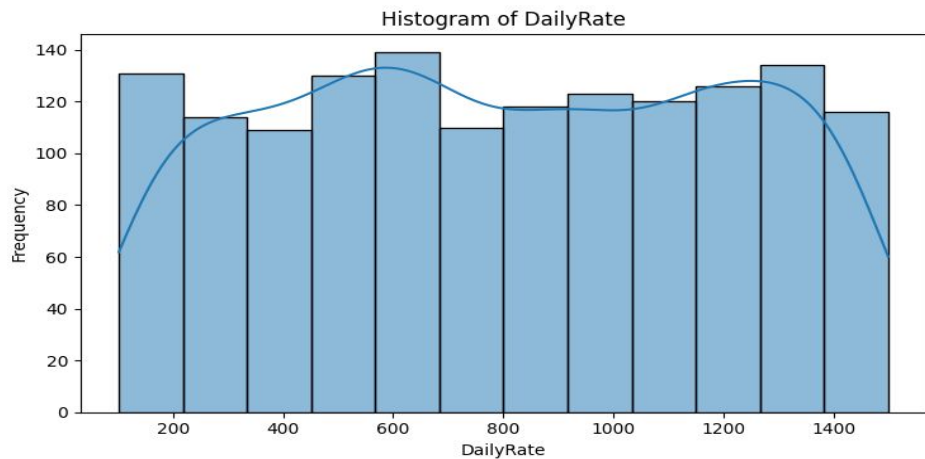
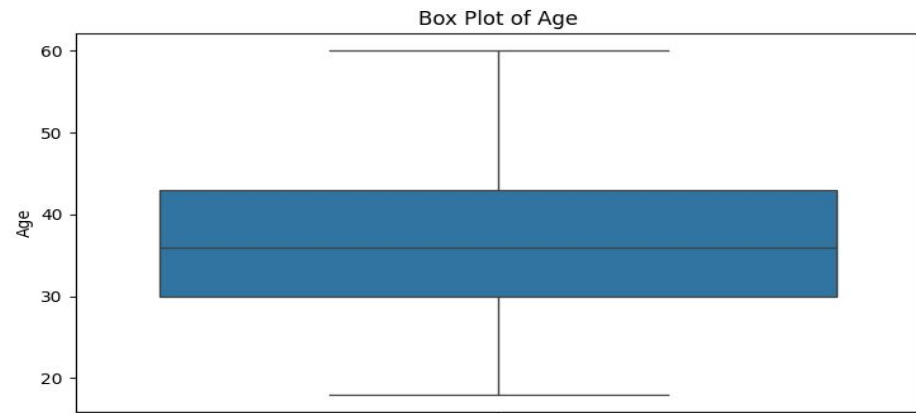
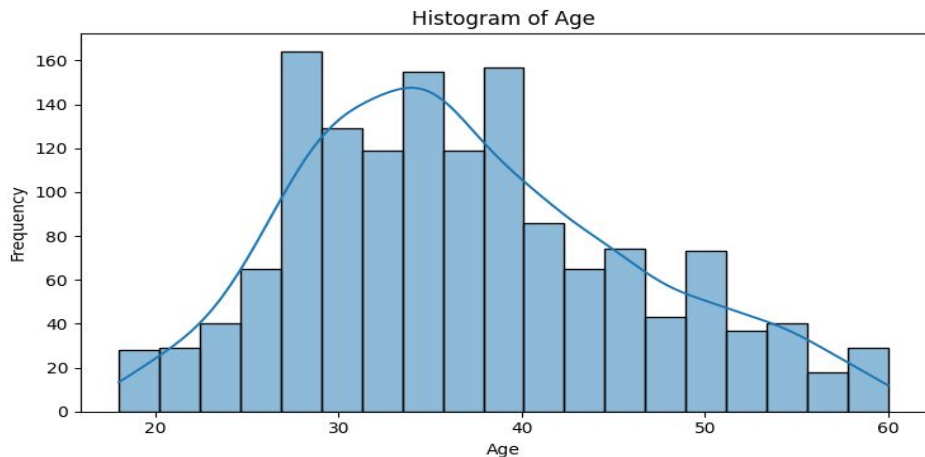
Distribution of Department (Sales)



Distribution of Marital Status (Single)

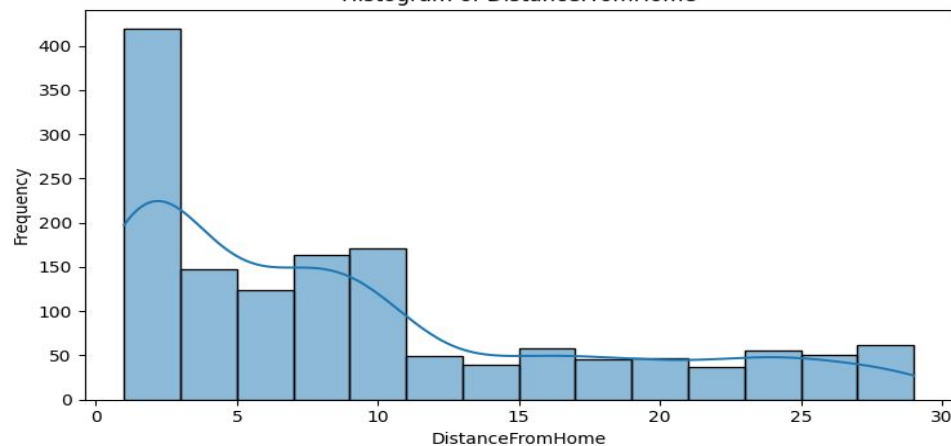


Distribution of Numerical Feature Batch 1

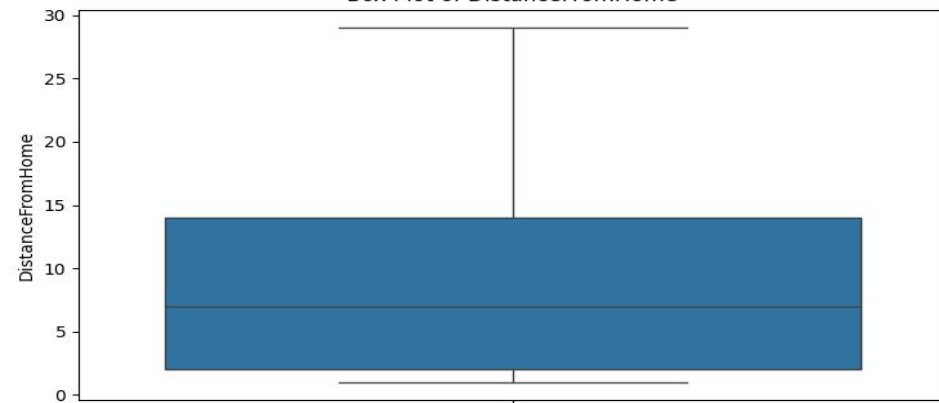


Distribution of Numerical Feature Batch 2

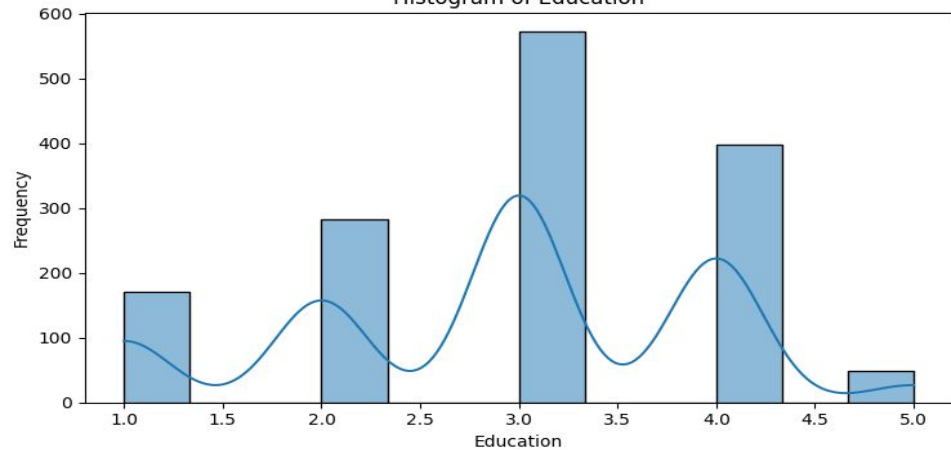
Histogram of DistanceFromHome



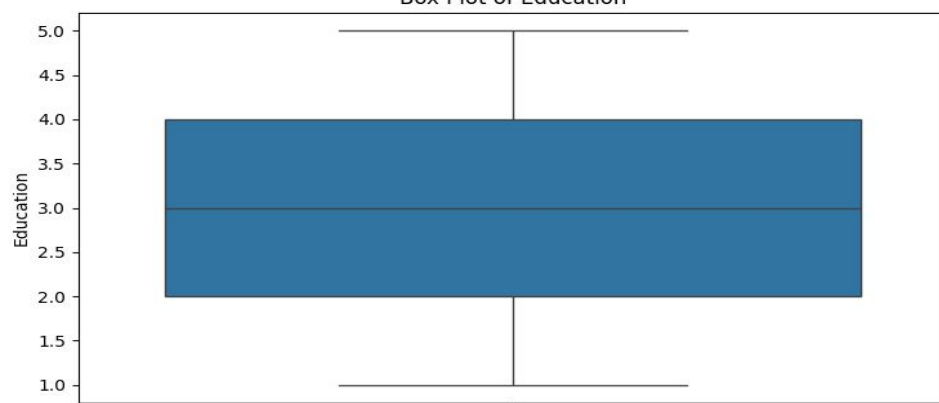
Box Plot of DistanceFromHome



Histogram of Education

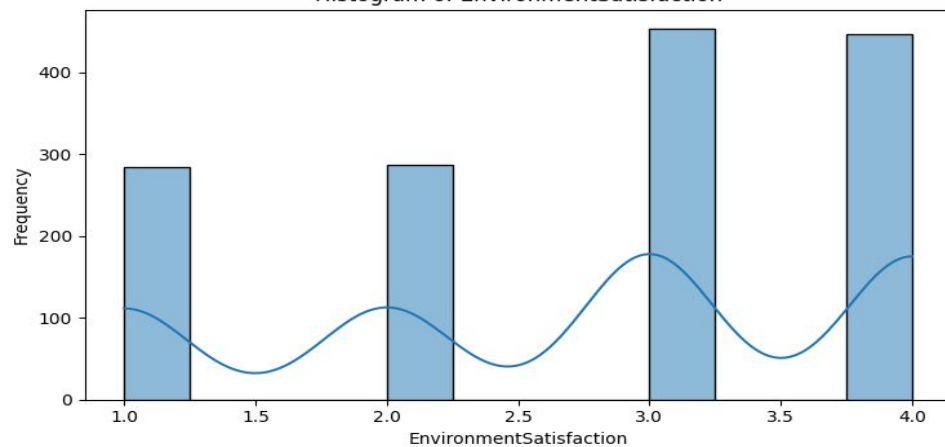


Box Plot of Education

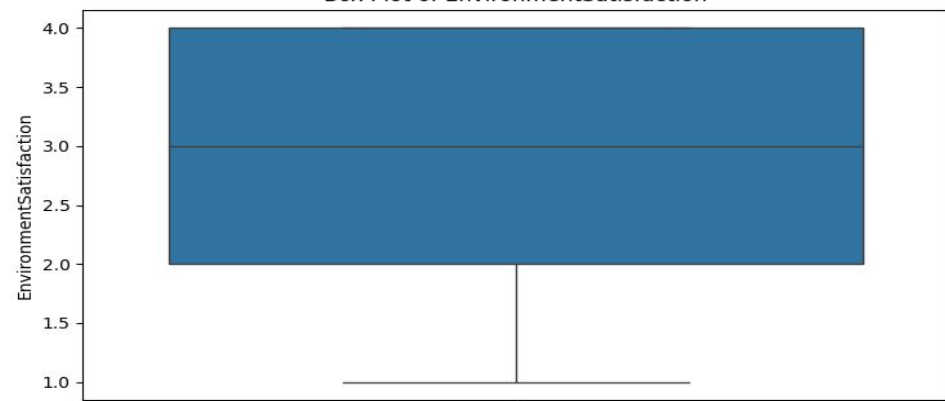


Distribution of Numerical Feature Batch 3

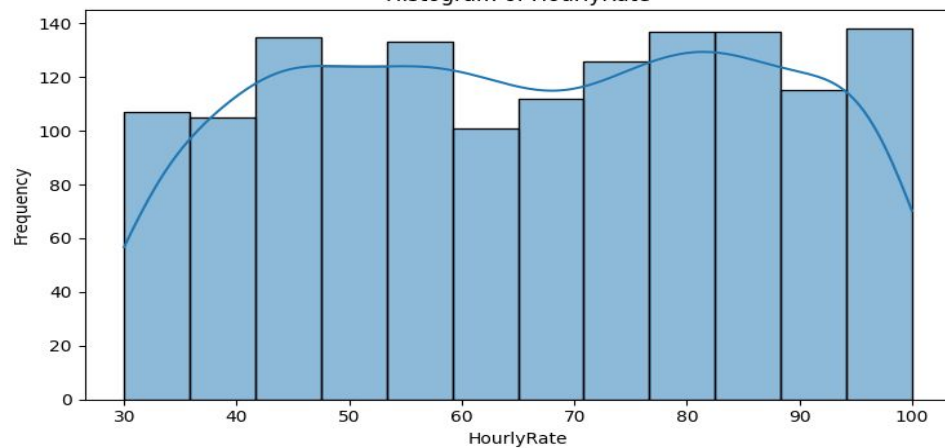
Histogram of EnvironmentSatisfaction



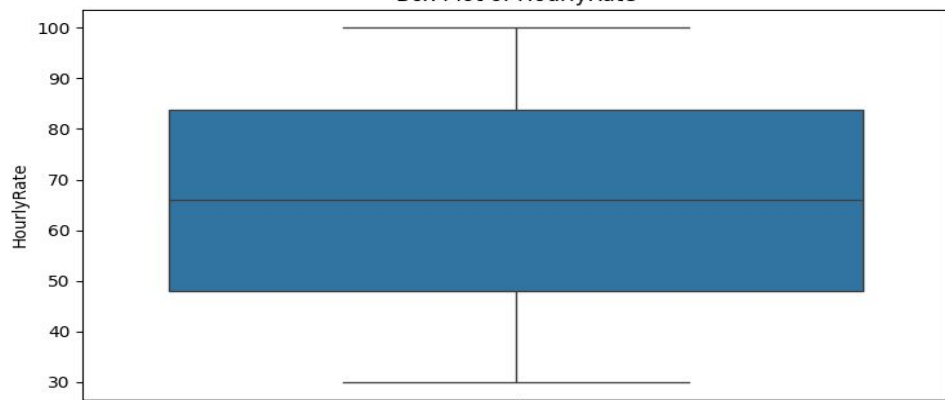
Box Plot of EnvironmentSatisfaction



Histogram of HourlyRate

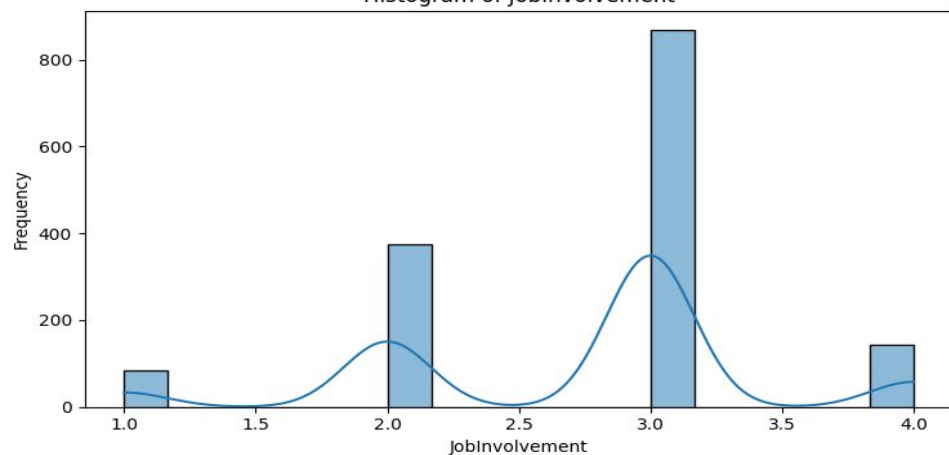


Box Plot of HourlyRate



Distribution of Numerical Feature Batch 4

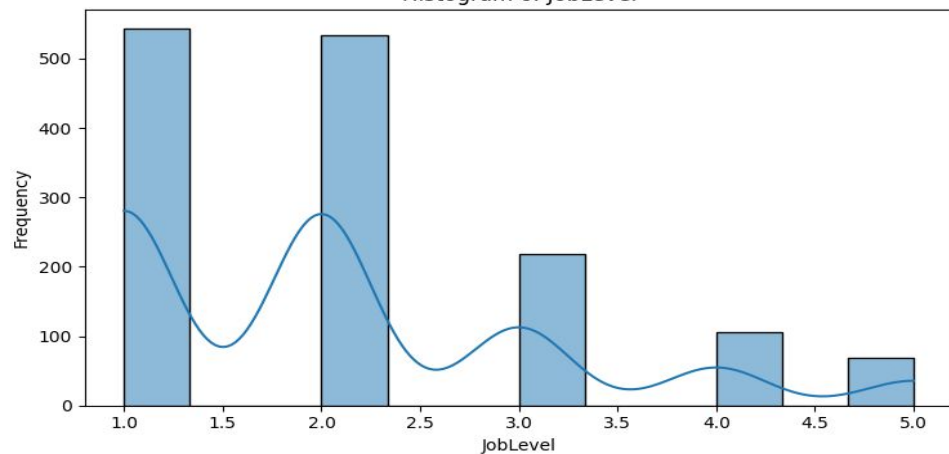
Histogram of JobInvolvement



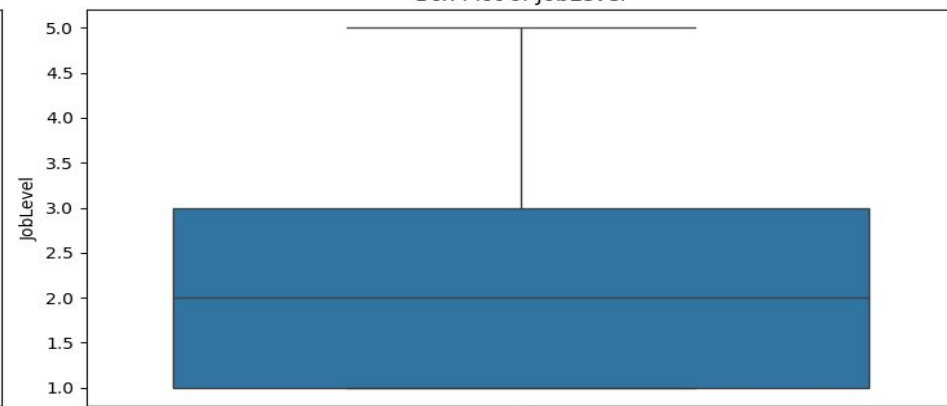
Box Plot of JobInvolvement



Histogram of JobLevel

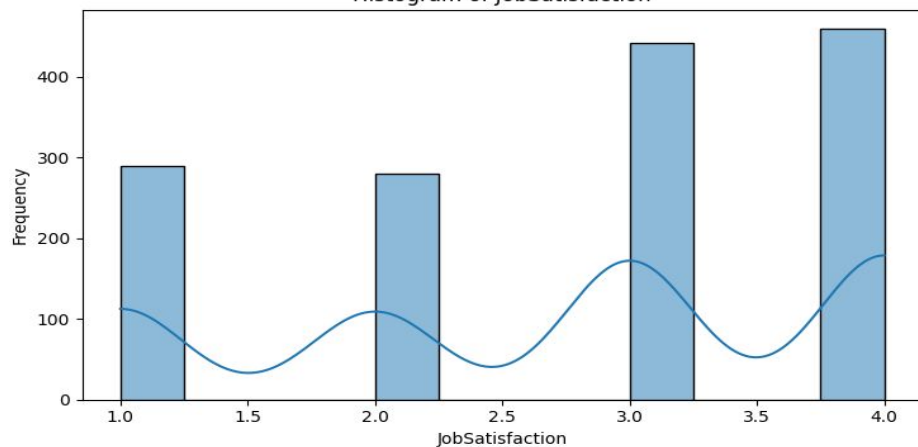


Box Plot of JobLevel

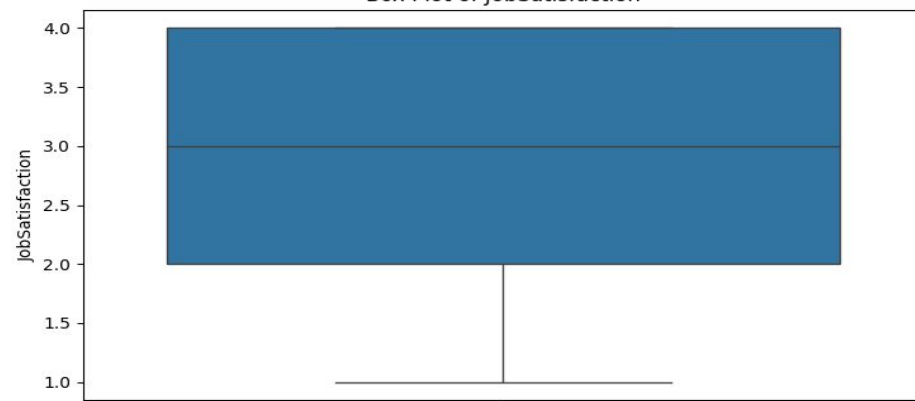


Distribution of Numerical Feature Batch 5

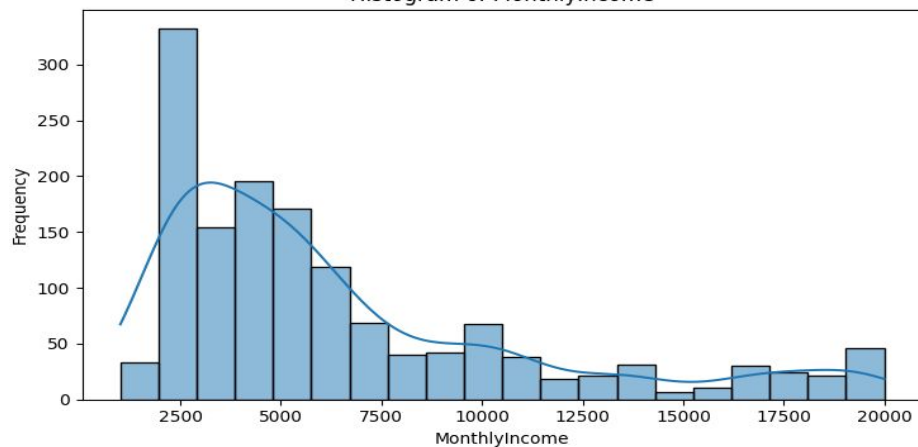
Histogram of JobSatisfaction



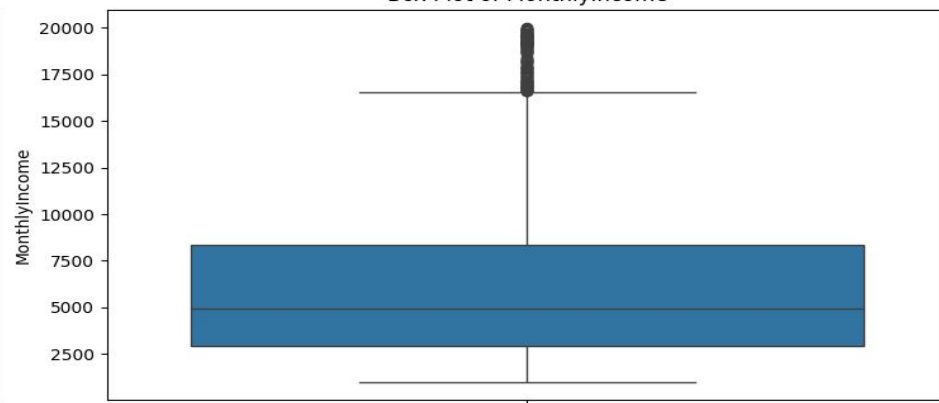
Box Plot of JobSatisfaction



Histogram of MonthlyIncome

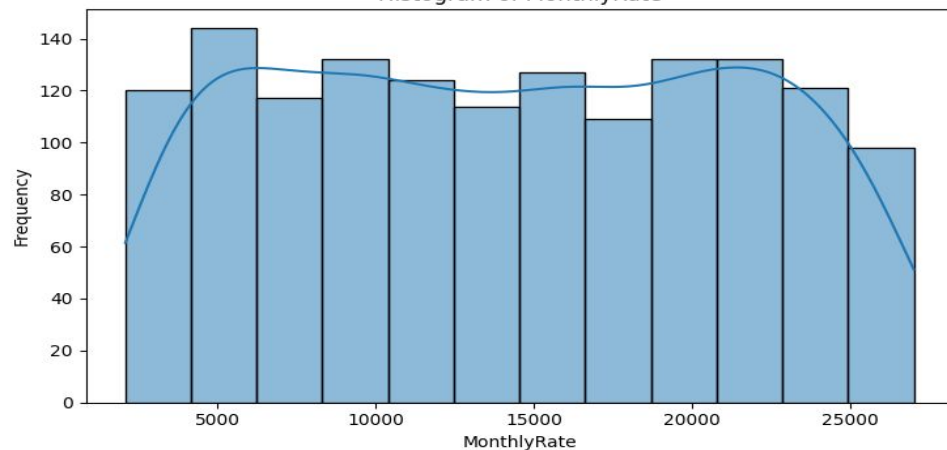


Box Plot of MonthlyIncome

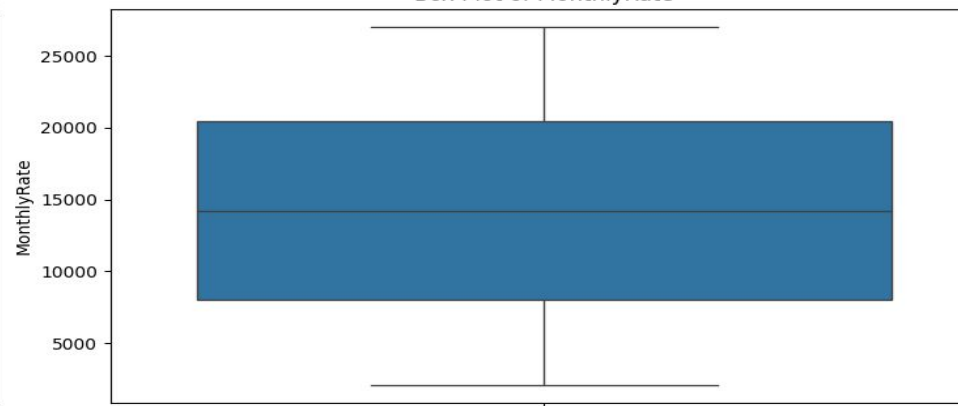


Distribution of Numerical Feature Batch 6

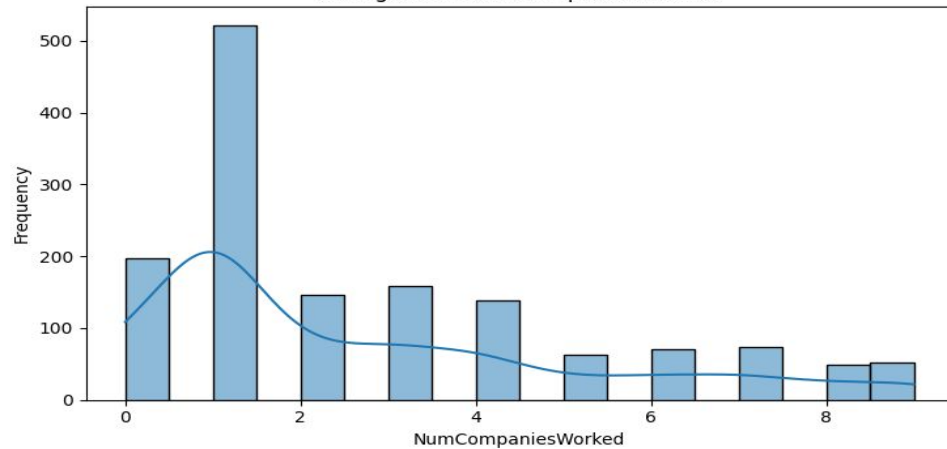
Histogram of MonthlyRate



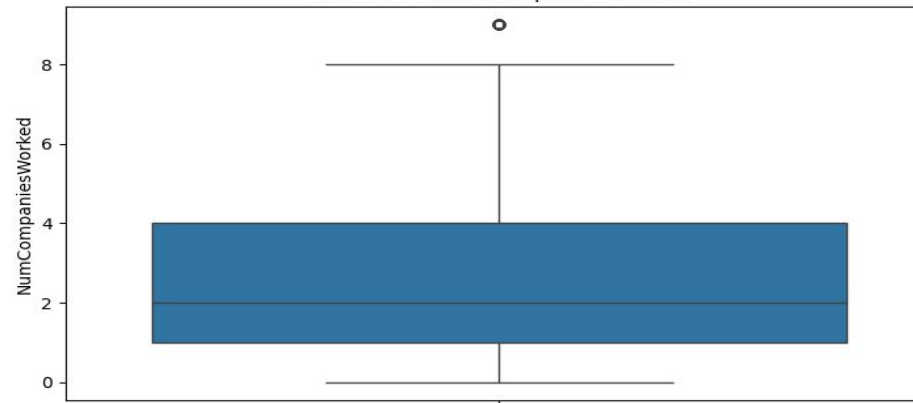
Box Plot of MonthlyRate



Histogram of NumCompaniesWorked

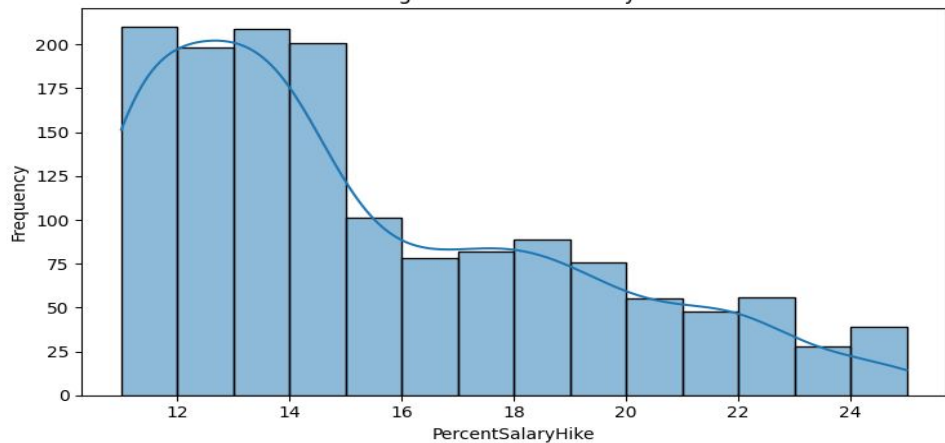


Box Plot of NumCompaniesWorked

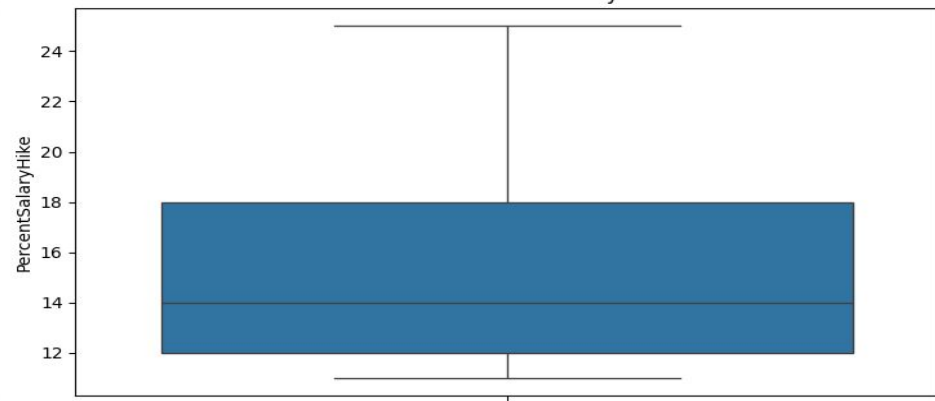


Distribution of Numerical Feature Batch 7

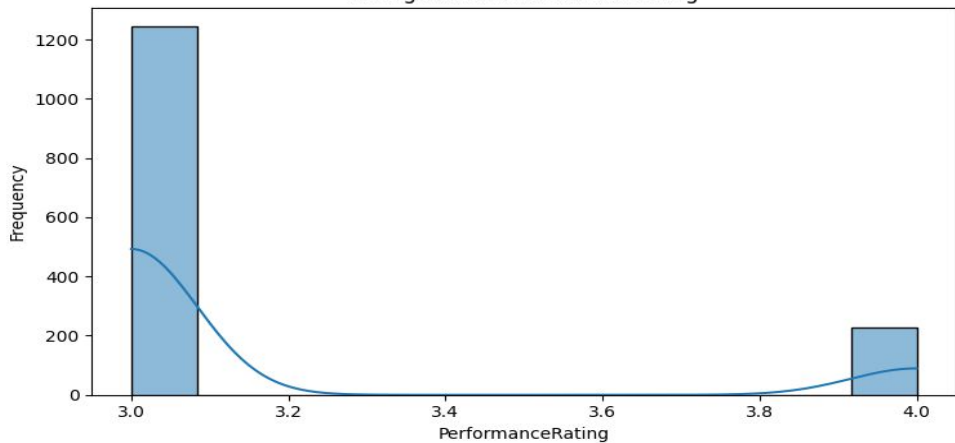
Histogram of PercentSalaryHike



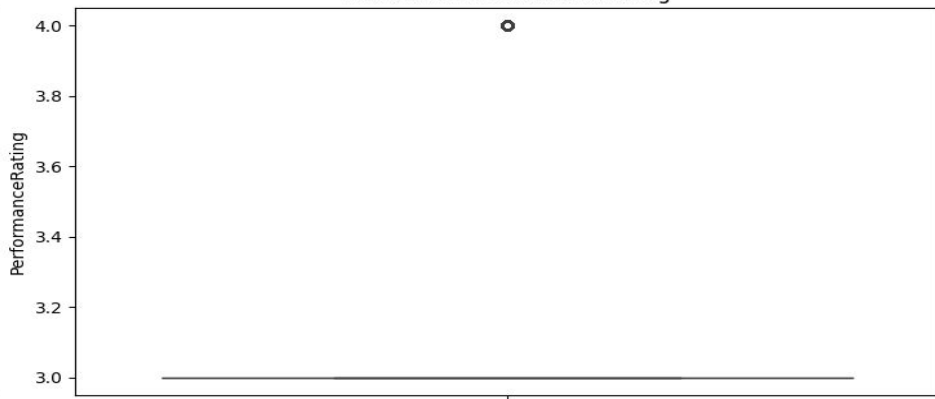
Box Plot of PercentSalaryHike



Histogram of PerformanceRating

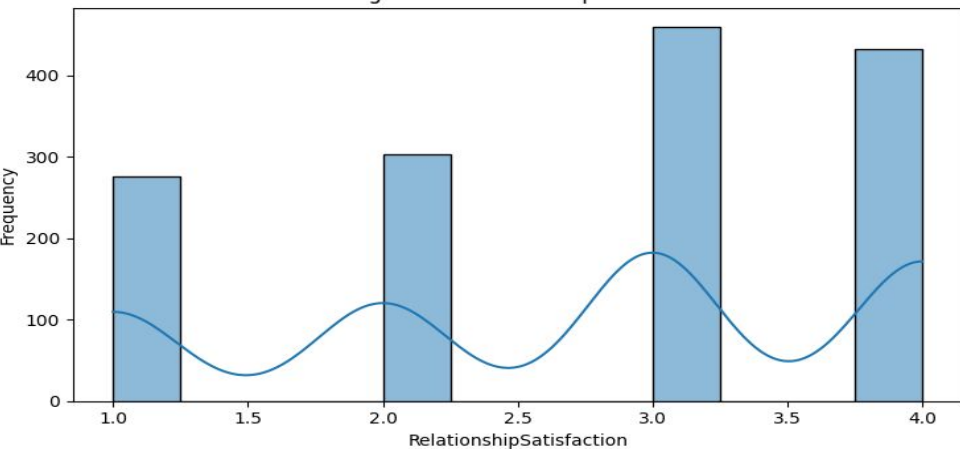


Box Plot of PerformanceRating

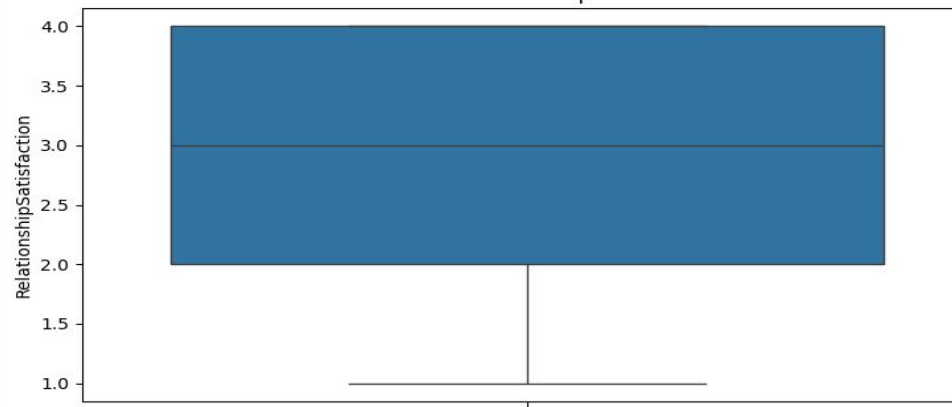


Distribution of Numerical Feature Batch 8

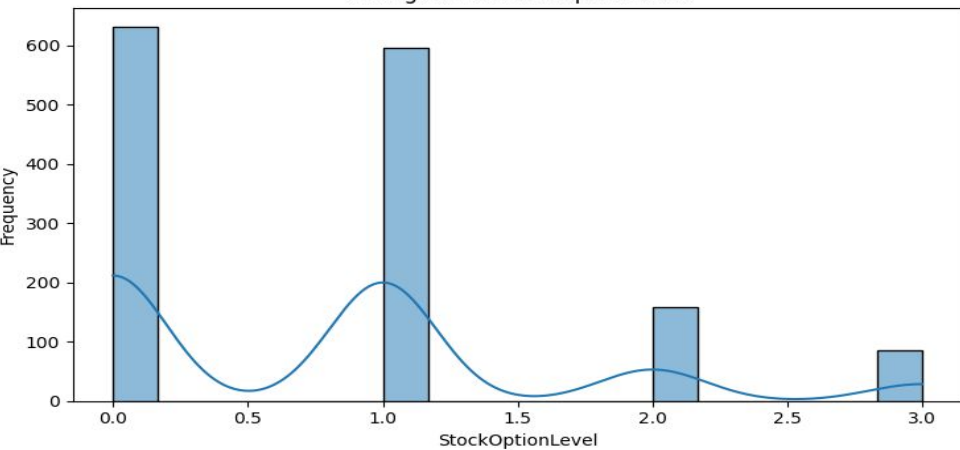
Histogram of RelationshipSatisfaction



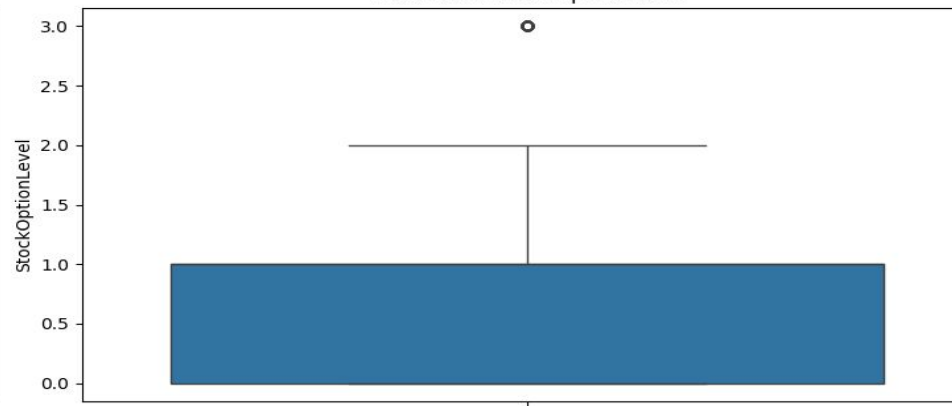
Box Plot of RelationshipSatisfaction



Histogram of StockOptionLevel

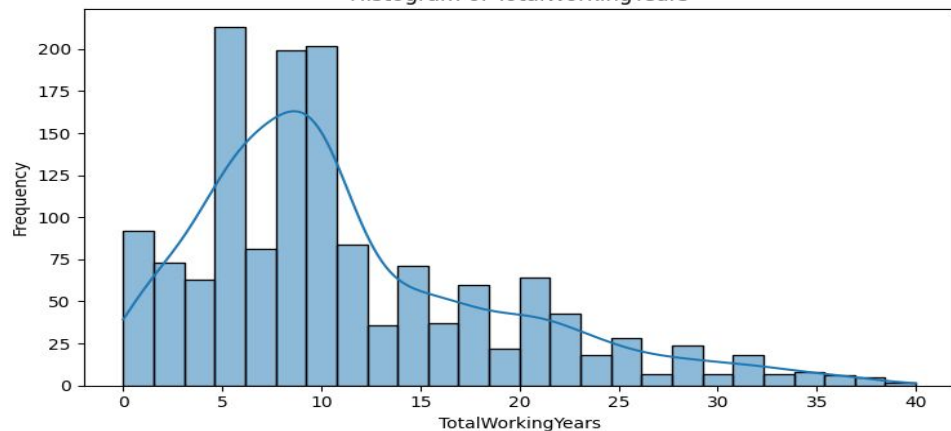


Box Plot of StockOptionLevel

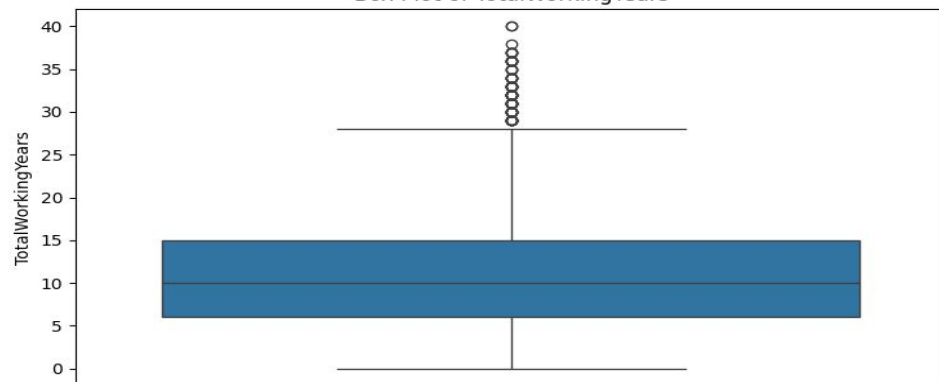


Distribution of Numerical Feature Batch 9

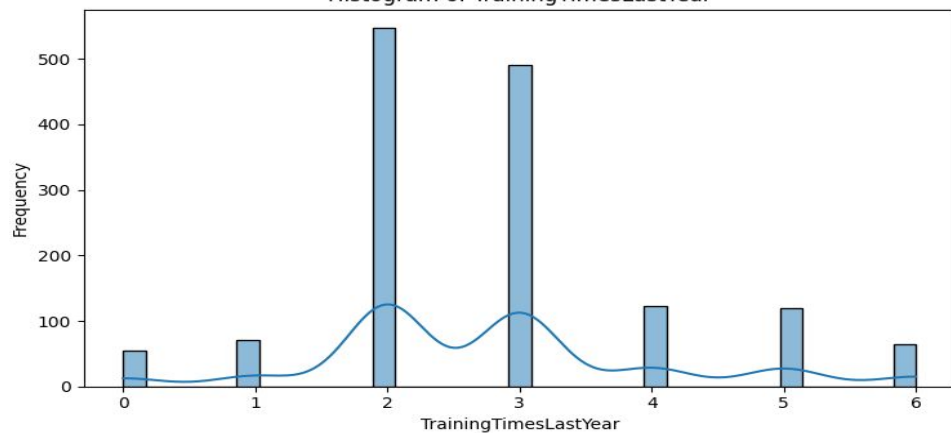
Histogram of TotalWorkingYears



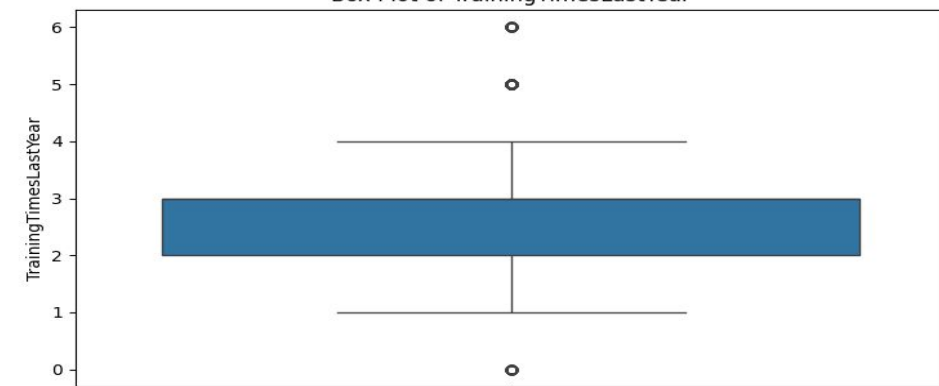
Box Plot of TotalWorkingYears

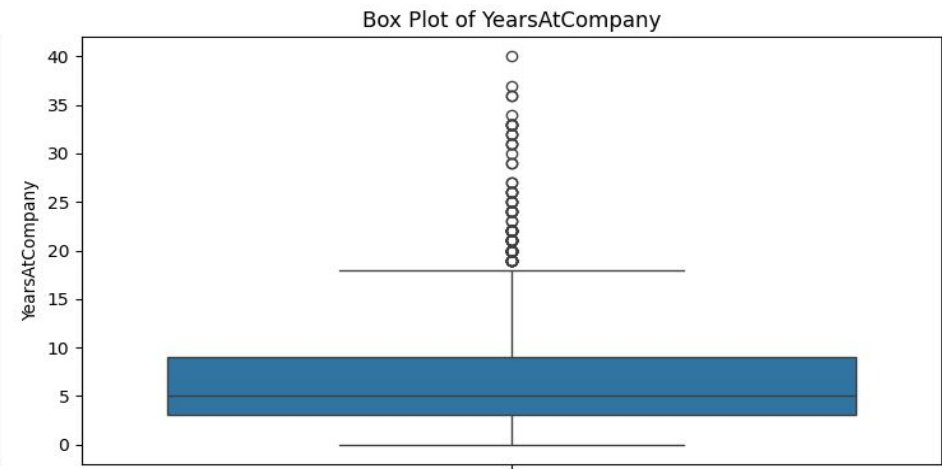
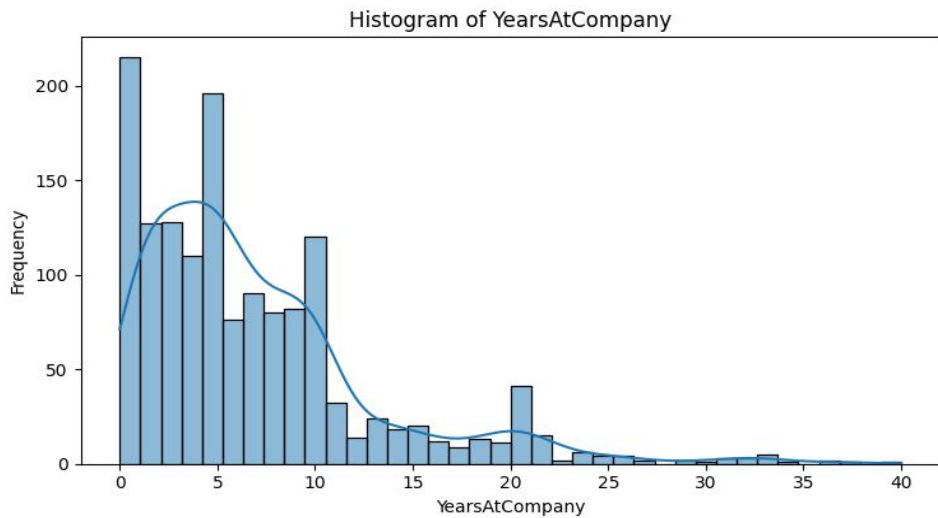
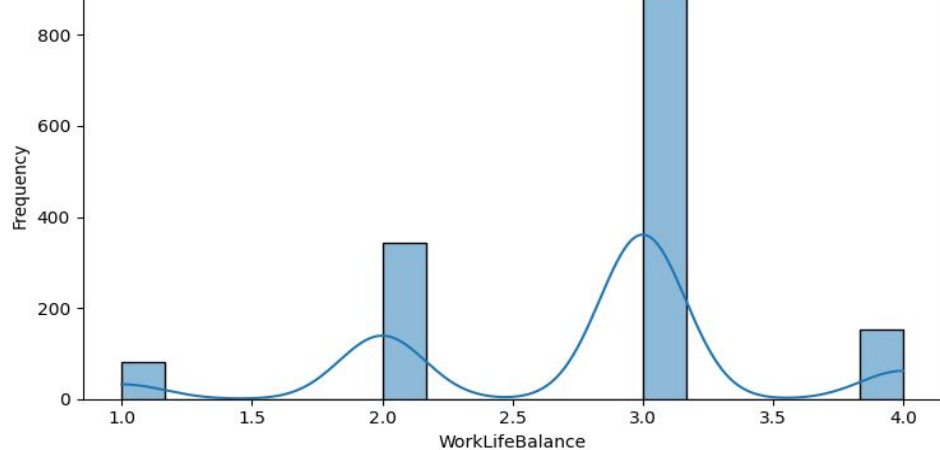


Histogram of TrainingTimesLastYear



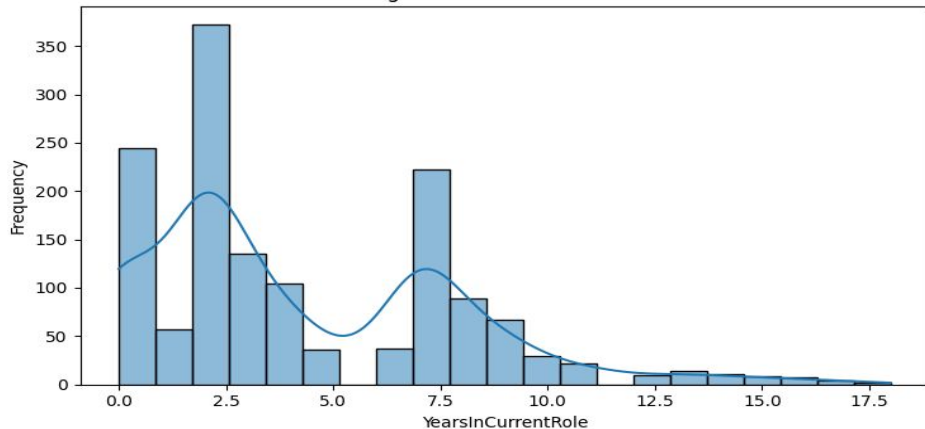
Box Plot of TrainingTimesLastYear



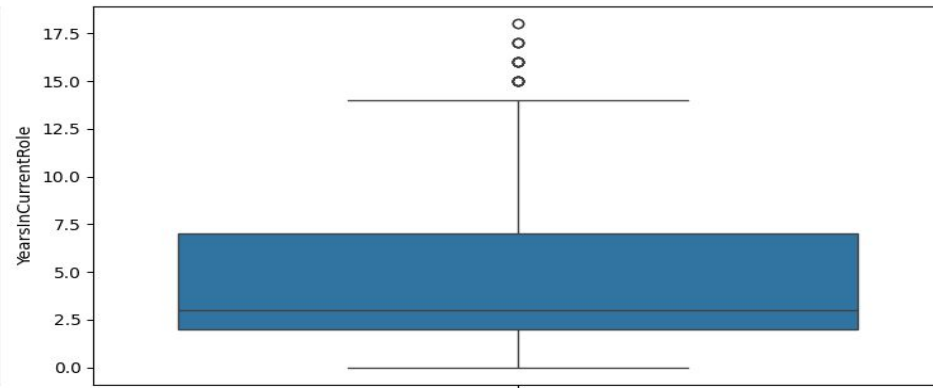


Distribution of Numerical Feature Batch 11

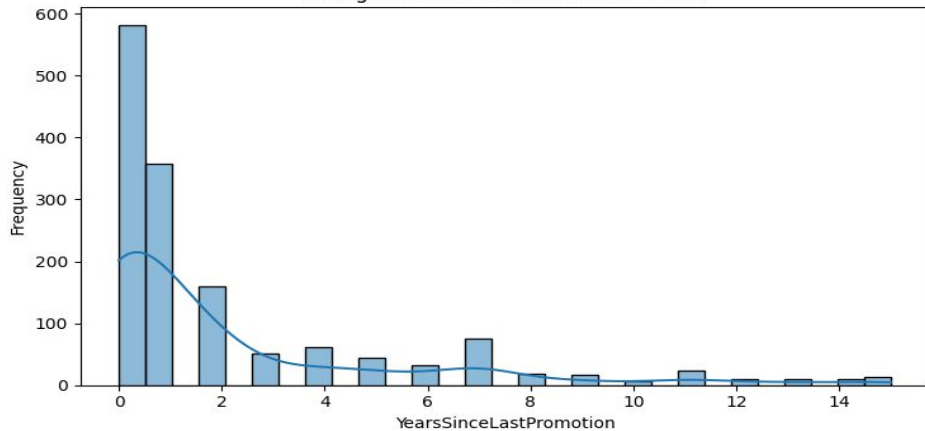
Histogram of YearsInCurrentRole



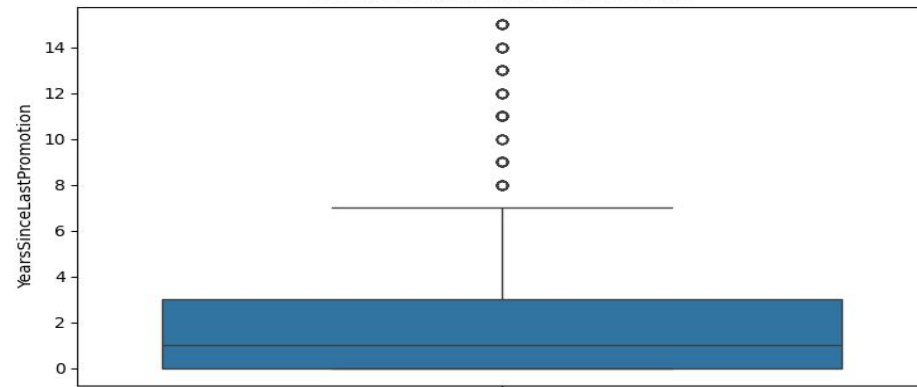
Box Plot of YearsInCurrentRole



Histogram of YearsSinceLastPromotion

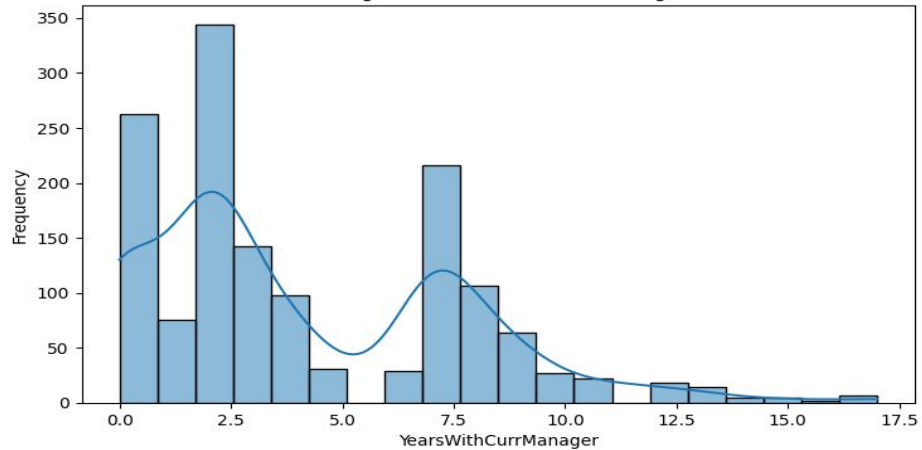


Box Plot of YearsSinceLastPromotion

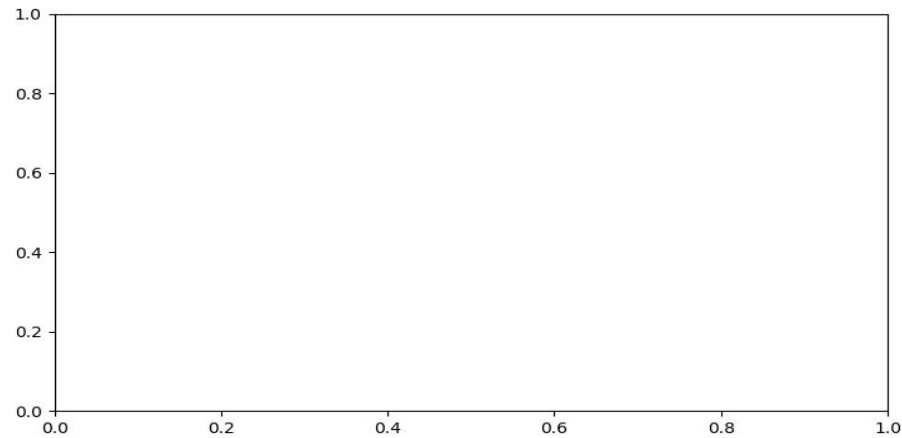
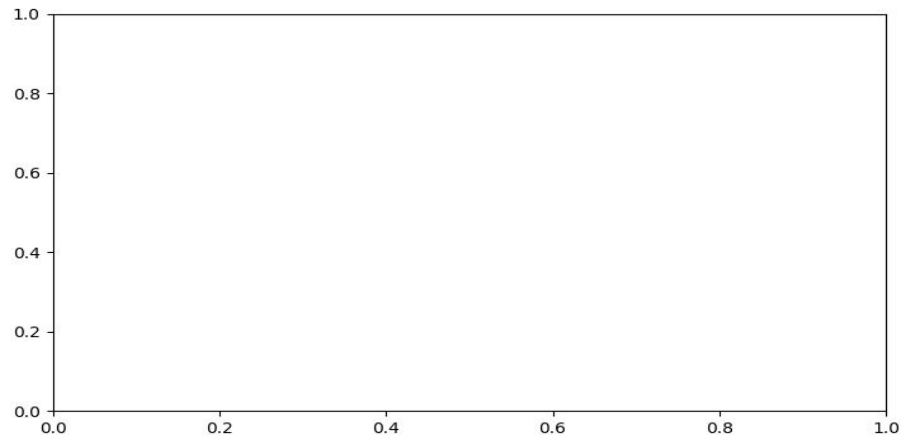
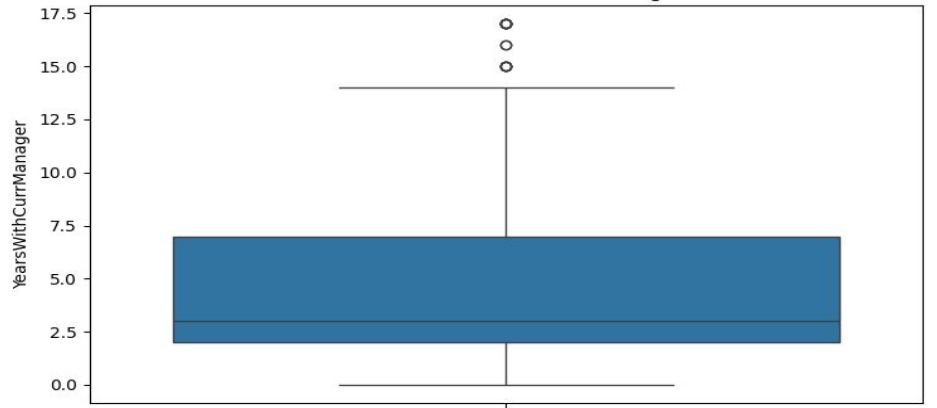


Distribution of Numerical Feature Batch 12

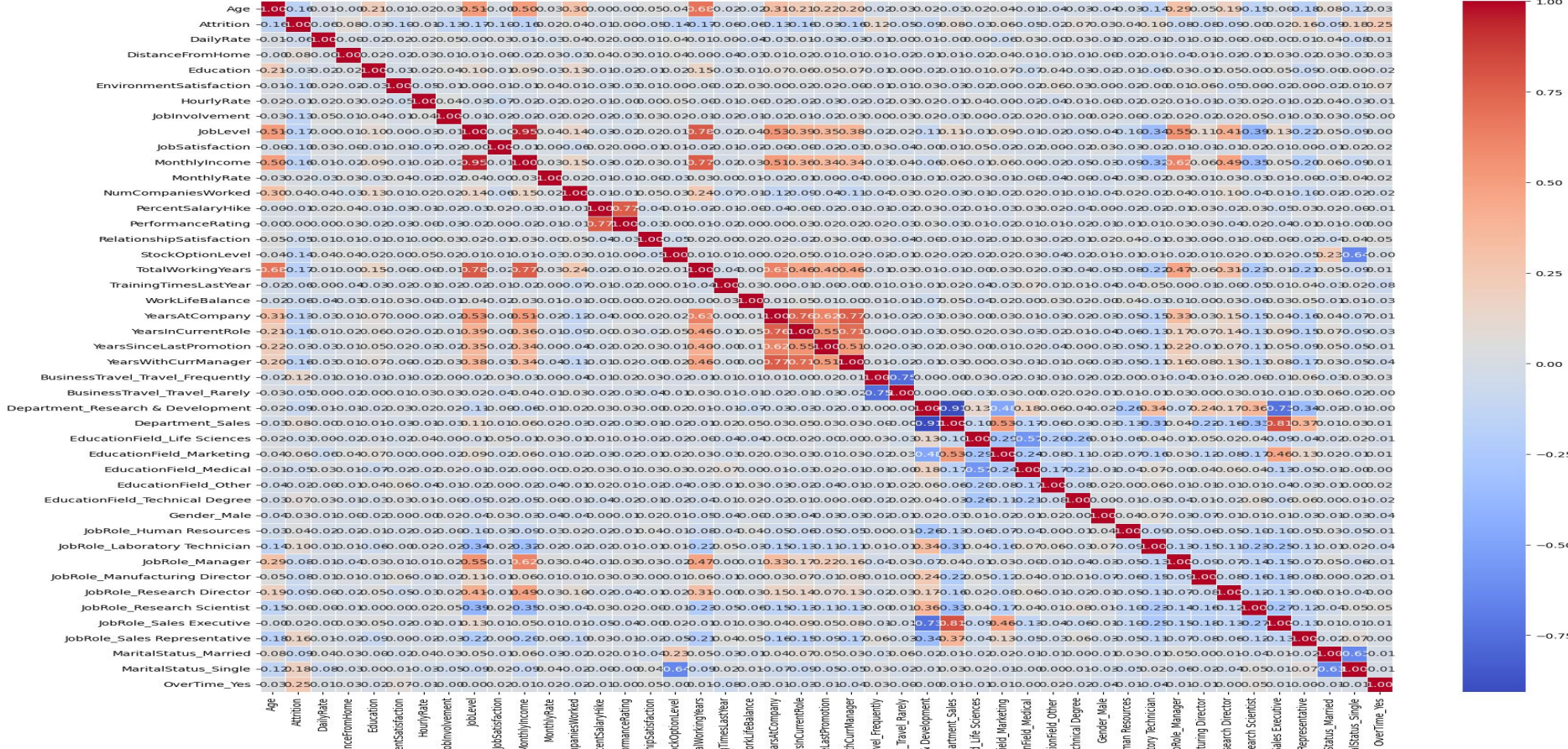
Histogram of YearsWithCurrManager



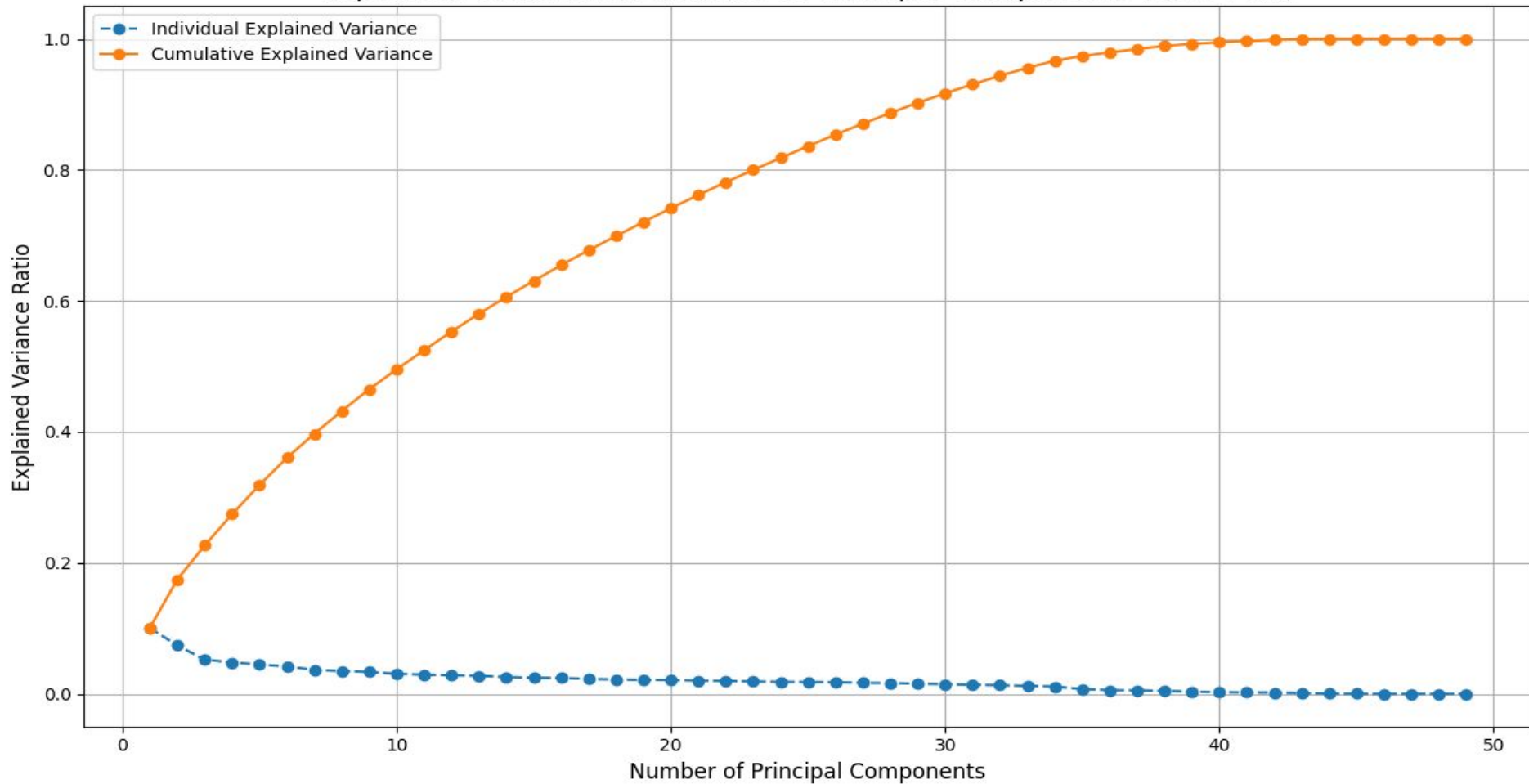
Box Plot of YearsWithCurrManager



Correlation Matrix of Numerical Features



Explained Variance vs. Number of Principal Components (New PCA)



Logistic Regression model :

Model Performance:

Accuracy: 0.8476

Precision: 0.5000

Recall: 0.4688

F1-Score: 0.4839

AUC-ROC: 0.7726

Beginning Decision Tree Classifier

Decision Tree Classifier -

Decision Tree Model Performance:

Accuracy: 0.7810

Precision: 0.3250

Recall: 0.4062

F1-Score: 0.3611

AUC-ROC: 0.6273

Beginning Random Forest Classifier

Random Forest Classifier

Random Forest Model

Performance:

Accuracy: 0.8524

Precision: 0.6000

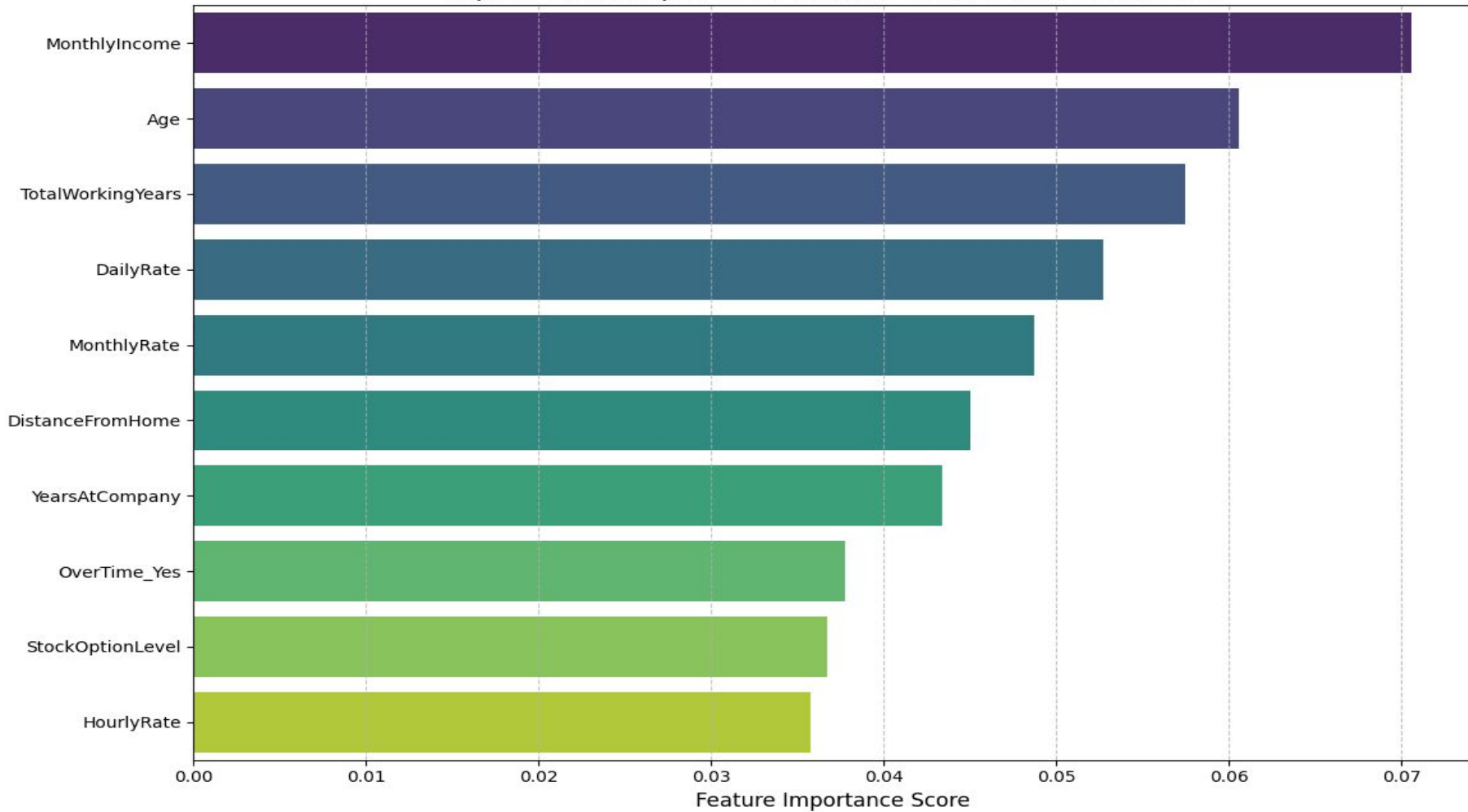
Recall: 0.0938

F1-Score: 0.1622

AUC-ROC: 0.6676

Top 10 Most Important Features for Attrition Prediction

Feature Name



⚙️ Settings

Choose Prediction Model:

Logistic Regression ▾

Logistic Regression selected.

Model Accuracy

87.41%

Go to Page:

- Attrition Prediction
- High-Risk Employee List
- Job Satisfaction & Performance
- Side-by-Side Comparisons



Attrition Risk Predictor

Currently using **Logistic Regression** to predict risk.

Input Details

Age



Monthly Income (\$)

5000 - +

Overtime

Yes ▾

Job Satisfaction



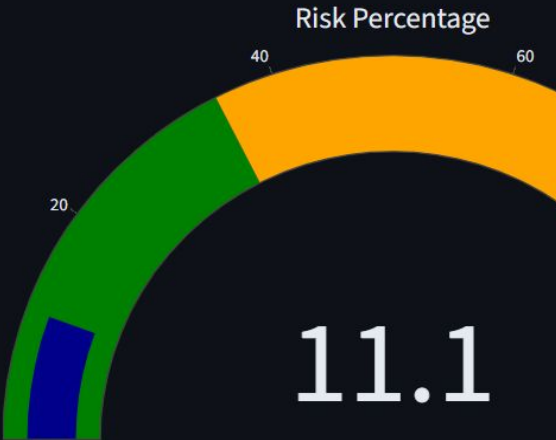
Environment Satisfaction



Distance From Home (km)



Results



Thank you